

Project Proposal:

Weather Prediction using Recurrent Neural Networks (RNNs)

1. Objective

The goal of this project is to build a deep learning model based on Recurrent Neural Networks (RNNs) for short-term weather prediction using time-series climate data. The project aims to explore how sequential models like LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit) can capture temporal patterns in historical weather observations to forecast future temperature, humidity, and precipitation.

2. Motivation

Accurate weather prediction is crucial for agriculture, disaster management, transportation, and daily life planning. Traditional methods, while effective, often rely on complex physical simulations. Deep learning offers a data-driven alternative, capable of capturing nonlinear dependencies in atmospheric variables. This project aligns with the course's focus on practical applications of deep learning in climate science.

3. Dataset

The model will be trained on publicly available weather datasets, such as:

- **NOAA Global Historical Climatology Network (GHCN)**
- **ERA5 reanalysis data (from Copernicus Climate Data Store)**
The dataset will be preprocessed into sequences of features like temperature, pressure, humidity, wind speed, and precipitation, with corresponding target variables for prediction.

4. Methodology

- **Preprocessing:** Handling missing values, normalizing features, creating time-series windows.
- **Modeling:**
 - Implement baseline RNN models.
 - Extend to LSTM and GRU variants.
 - Compare performance of different architectures.
- **Training & Evaluation:**

5. Expected Outcomes

- A trained deep learning model capable of predicting short-term weather variables based on historical data.
- Insights into how RNN-based architectures perform on real-world climate data.
- Comparative evaluation between RNN, LSTM, and GRU models in the context of climate prediction.